

Phytochemicals and their impact on adipose tissue inflammation and diabetes.



Type 2 diabetes mellitus is an inflammatory disease and the mechanisms that underlie this disease, although still incompletely understood, take place in the adipose tissue of obese subjects. Concurrently, the prevalence of obesity caused by Western diet's excessive energy intake and the lack of exercise escalates, and is believed to be causative for the chronic inflammatory state in adipose tissue. Overnutrition itself as an overload of energy may induce the adipocytes to secrete chemokines activating and attracting immune cells to adipose tissue. But also inflammation-mediating food ingredients like saturated fatty acids are believed to directly initiate the inflammatory cascade. In addition, hypoxia in adipose tissue as a direct consequence of obesity, and its effect on gene expression in adipocytes and surrounding cells in fat tissue of obese subjects appears to play a central role in this inflammatory response too. In contrast, revisiting diet all over the world, there are also some natural food products and beverages which are associated with curative effects on human health. Several natural compounds known as spices such as curcumin, capsaicin, and gingerol, or secondary plant metabolites catechin, resveratrol, genistein, and quercetin have been reported to provide an improved health status to their consumers, especially with regard to diabetes, and therefore have been investigated for their anti-inflammatory effect. In this review, we will give an overview about these phytochemicals and their role to interfere with inflammatory cascades in adipose tissue and their potential for fighting against inflammatory diseases like diabetes as investigated in vivo. Copyright © 2012 Elsevier Inc. All rights reserved.